EXISTENCE OF A NONZERO C WITH A/CA REDUCED AND A DIVIDING A POWER OF C IN A TWO-DIMENSIONAL NORMAL LOCAL DOMAIN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- There exists c in the intersection of the height-one primes p_i such that A/cA is reduced. invoked at proof lines 38249-38250. Provides the element c with the required reduced quotient.
- In a normal domain, an element x lies in A as soon as it lies in each localization at a height-one prime. invoked at proof lines 38253-38254. Converts the effectivity of the divisor into the containment c^n/a in A.

Source reference. Original lemma 0AXH: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38138–38138. Cited at proof lines 38248-38250 for the notation of the divisor and the choice of c; the conclusion is used inside the counted span.

Source reference. Original lemma 031T: sources/stacks-chow/tags.txt, source lines 3996–3996. Cited at proof lines 38253-38254 to conclude c^n/a lies in A from its presence in every height-one localization; the conclusion is used inside the counted span.

Lemma 0.2. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian normal local domain of dimension 2. If $a \in \mathfrak{m}$ is nonzero, then there exists an element $c \in A$ such that A/cA is reduced and such that a divides c^n for some n.*

Proof. Let $\mathrm{div}(a) = \sum_{i=1,\ldots,r} n_i \mathfrak{p}_i$ with notation as in the proof of Lemma 0AXH. Choose $c \in \mathfrak{p}_1 \cap \ldots \cap \mathfrak{p}_r$ with A/cA reduced, see Lemma 0AXH. For $n \geq \max(n_i)$ we see that $-\mathrm{div}(a) + \mathrm{div}(c^n)$ is an effective divisor (all coefficients nonnegative). Thus $c^n/a \in A$ by Algebra, Lemma 031T. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.